AMIGA

Experimental design: five phases

Study 1 · labelled benchmark evaluation

1 2 3 4 5

Choose how to rank. Test on held-out topologies. Vary the amount of labelled data.

Two generators, the same benchmark problems

87 topology groups
All conditions of a topology stay together

- **BIO-INSIGHT** — 104 fronts
- **MO-GENECI** — 104 fronts

Analyze each generator separately, not as 208 independent networks.

Keep topology groups together across nested splits

5 outer folds

Outer training topologies
3 inner folds for phases 1–3

Outer test topologies
Reserved for phases 4–5

Repeat the complete protocol for every outer split.

DEVELOP THE PROCEDURE — Training + inner validation only

Outer test data stay untouched

1 Choose the relevance labels

Translate known network quality into learning targets.

Known quality within a front → AUPR

Rank order	Relative quality	Quality bins
4	1.00	high
3	0.75	high
2	0.50	mid
1	0.25	low
0	0.00	low
dense / average	0–1 per front	5 / 10 / 15 bins

↓↑ Also test reversed + shuffled labels (controls)

LightGBM · XGBoost · CatBoost

Keep one label scheme per learner

Regression and classification use fixed targets.

2 Tune each learner

Try predefined parameter combinations.

LightGBM	XGBoost	CatBoost
27 settings tested	36 settings tested	36 settings tested

Score each setting on 3 inner folds

Keep parameters for each learner × task

Same search policy for each supervised formulation.

3 Choose descriptors + learner

Remove less useful predictors using training data.

Training-only importance for candidate ordering (TreeSHAP)

All columns → 75% → 50% → 25%

Columns retained from the original table
Refit after each reduction
Inner validation chooses the retained fraction and the model family.

Keep a feature budget + learner

Relearn the selected column mask before outer evaluation.

Freeze the procedure for this outer split
Model family · labels · parameters · feature budget

Selection rule: minimize missed quality among the first 5 recommendations
Regret@5 → Regret@1 → fewer columns in phase 3 → stable tie-break

EVALUATE FIXED PROCEDURES

No test-driven selection

4 Compare on held-out topologies

Refit selected procedures; predict before reading test quality labels.

Same held-out candidates

Five supervised selectors

- **AMIGA · learn an order**
- Regression · predict AUPR
- Classification · top 5%
- Classification · top 10%
- Classification · top 20%

Assess the first 5

Best in the front
Best among top 5

Missed quality = Regret@5

Separate panel: objective-only rules + exact random reference
The AUPR oracle is a nondeployable ceiling, not a competitor.

5 final training seeds

Equal weight per topology
Regret@5 primary · Friedman + Holm, per generator

No seed selection or prediction ensemble; all five final seeds contribute.

5 Learn with fewer labelled topologies

Keep the AMIGA recipe and exact feature columns from phase 4.

Fixed model family · labels · parameters · exact columns

Labelled training topology groups

10 → 20 → 40

Full training set
Reuse phase-4 predictions
69–70

3 nested subset draws × 5 seeds · unchanged test folds

Measure learning curves
Regret@5 and Hit@5

Conditional 95% intervals from topology bootstrap

AMIGA only · no new model or column selection

10 · 20 · 40 · full

Conditional on the recipe chosen with the full training complement.

PARTITION
Whole topologies stay together

MEASURE
Regret: quality missed in the top k

AGGREGATE
Phase 4: seeds → fronts → topologies
Phase 5: seeds → subsets → fronts → topologies

INTERPRET
Internal evaluation on reused benchmarks

Study 2 (TCGA-BRCA) and prediction costs are separate analyses, not extra phases here.

Design schematic · icons and bars are illustrative, not experimental results.